Analysis of Motor Vehicle Crash Severity in New York City (2023–2024)

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Table of contents

Title Page…… 1

Table of Contents…… 2

Abstract……...……. 3

Introduction/Motivation…...……. 3

Data Set…… 4

Analysis…… 4

Results……. 6

Results Discussion …… 10

Key Analytical Insights……. 10

Policy and Practical Implications…….…. 11

Limitations…… 11

Conclusion…… 12

References…… 13

Abstract

Motor vehicle crashes represent a significant public safety challenge in large metropolitan areas such as New York City. Understanding the patterns and factors contributing to crash severity is critical for improving transportation safety policies and reducing injuries and fatalities. This study analyzes crash severity patterns using the NYC Motor Vehicle Collisions dataset obtained from the New York City Open Data Portal.

Crash severity is defined as the total number of injuries and fatalities per collision. Using crash data from January 2023 through June 2024, this study applies descriptive analytics, geographic visualization, and temporal analysis using Tableau to explore patterns related to crash severity. The dataset includes 144,863 crashes resulting in 82,179 injuries and 401 fatalities, with an average crash severity of 0.5701 persons injured or killed per collision. The analysis reveals that crash severity varies geographically across boroughs, temporally across hours and months, and behaviorally based on driver actions and vehicle types.

The findings indicate that driver behavior, commuting patterns, and vulnerable vehicle types significantly influence crash severity. These insights demonstrate how data visualization tools can be used to identify patterns in large datasets and support evidence-based transportation safety strategies.

Introduction/Motivation

Urban transportation systems involve complex interactions among vehicles, pedestrians, cyclists, and public transit systems. As cities become more densely populated, traffic congestion and roadway interactions increase, raising the risk of collisions.

New York City is one of the most densely populated urban environments in the world, with millions of daily commuters using a combination of personal vehicles, taxis, buses, bicycles, and pedestrian pathways. As a result, understanding traffic safety patterns in the city is essential for improving urban transportation planning and public safety.

This study focuses on identifying patterns in motor vehicle crash severity in New York City. Crash severity is measured using the number of injuries and fatalities associated with each collision.

The study addresses the following key research questions:

1. When and where do severe motor vehicle crashes most frequently occur in New York City?
2. How does crash severity vary across time of day, day of week, and month?
3. Which contributing factors and vehicle types are most strongly associated with severe crashes?

To answer these questions, the study uses descriptive analytics and interactive visualizations built in Tableau to explore crash data across geographic, temporal, and behavioral dimensions.

Traffic crashes remain one of the leading causes of injury and death in urban environments. According to the National Highway Traffic Safety Administration, thousands of fatalities occur annually due to roadway collisions in the United States. Understanding patterns of crash severity

is therefore essential for developing effective public safety policies. In densely populated cities such as New York City, where millions of daily transportation interactions occur, identifying high-risk behaviors, locations, and time periods can help policymakers prioritize safety interventions and reduce preventable injuries.

Data Set

The dataset used for this project is NYC Motor Vehicle Collisions – Crashes, obtained from the New York City Open Data portal.

Each record in the dataset represents a reported motor vehicle crash and includes detailed information about the time, location, contributing factors, and outcomes of the crash.

The dataset used in this analysis includes crash records from January 2023 through June 2024.

Summary statistics from the dataset include:

- Total crashes: 144,863
- Total injuries: 82,179
- Total fatalities: 401
- Average crash severity: 0.5701

These values represent the overall scale of traffic incidents in New York City during the study period.

Key variables used in the analysis include:

Variable	**Description**
Crash Date Time	Date and time when the crash occurred
Borough	Borough in which the crash occurred
Latitude and Longitude	Geographic coordinates of the crash
Persons Injured	Number of injuries reported
Persons Killed	Number of fatalities per crash
Contributing Factor	Reported cause of the crash
Vehicle Type	Type of vehicle involved in the crash

Before analysis, several preprocessing steps were conducted, including:

• Removal of missing geographic coordinates

• Creation of a crash severity metric - Crash severity was calculated using the formula:

Severity = Persons Injured + Persons Killed

• Extraction of hour, weekday, and month variables from the crash datetime field

• Aggregation of crash data by geographic and temporal dimensions

These steps allowed the dataset to be effectively visualized using Tableau dashboards.

Analysis

The analysis was conducted using Tableau and Excel to identify patterns in crash severity across geographic locations, time periods, and contributing factors.

The analytical process followed several stages:

Descriptive Analysis
Initial descriptive statistics were calculated to understand the scale of crash incidents in the dataset.

Geographic Analysis
Crash locations were mapped using latitude and longitude coordinates to identify clusters of collisions across New York City.

Temporal Analysis
Crash patterns were analyzed across several time dimensions including:
• Hour of day
• Day of week
• Month of year

Behavioral Analysis
Crash severity was examined across contributing factors and vehicle types to determine which conditions are associated with more severe collisions.

Predictive Modeling Considerations
Although the primary focus of this project is descriptive and exploratory analysis through data visualization, predictive modeling techniques could provide additional insights into crash severity patterns. Predictive analytics methods such as multiple linear regression or logistic regression could be used to examine how independent variables influence crash severity.
For example, variables such as borough, time of day, contributing factors, and vehicle type could be used as predictor variables to estimate the likelihood of injuries or fatalities in a crash. By modeling the relationship between these factors and crash severity, researchers could identify which conditions most strongly increase the risk of severe collisions.
Predictive models could also support proactive traffic safety strategies. Transportation planners could use such models to forecast high-risk crash conditions and allocate resources more effectively, such as increasing enforcement during peak crash periods or improving roadway infrastructure in high-risk locations.
Future research could expand this analysis by incorporating predictive modeling techniques using statistical software such as Excel regression tools, R, or Python machine learning libraries. Integrating predictive analytics with visualization tools like Tableau would allow policymakers to both understand current patterns and anticipate future traffic safety challenges.

Results

Overall Crash Metrics

Figure 1 – Figure 1. Total number of recorded motor vehicle crashes in New York City (2023–2024).

This visualization displays the total number of recorded crashes during the study period.

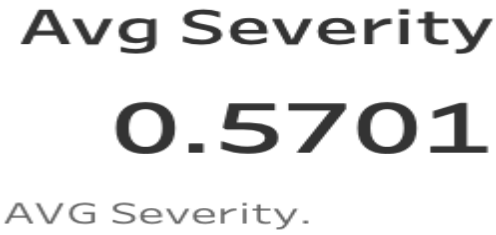
Figure 2 – Total Injuries KPI

This visualization shows the total number of individuals injured in motor vehicle collisions.

Figure 3 – Total Fatalities KPI

This visualization summarizes the total number of fatalities resulting from traffic collisions.

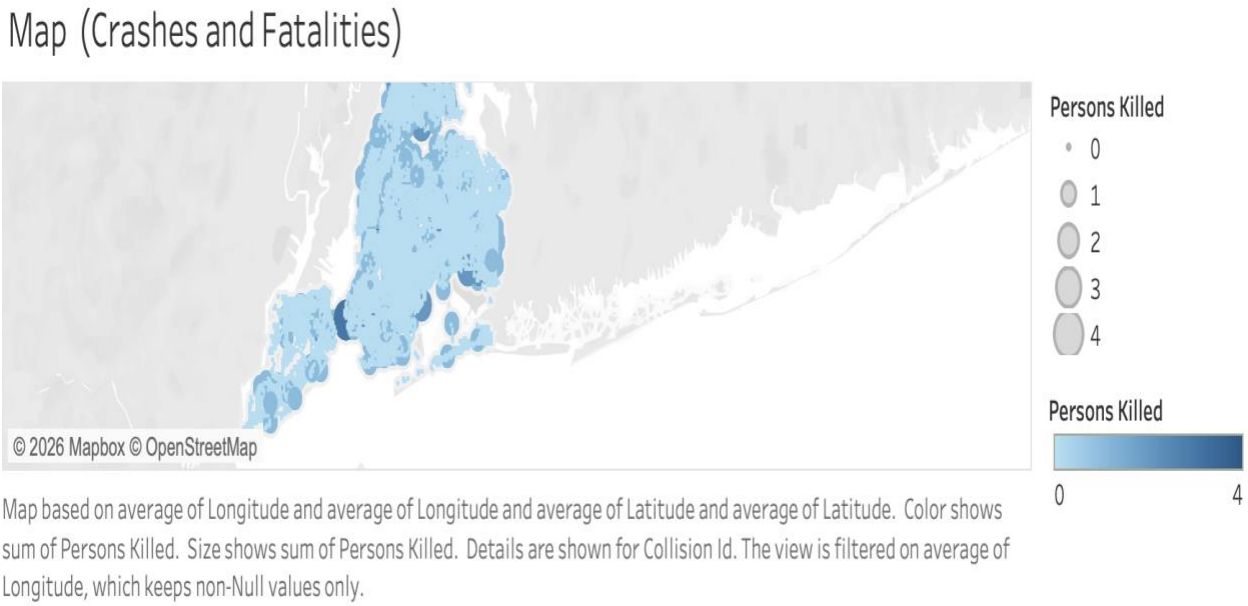
Figure 4 – Average Crash Severity



Average crash severity represents the average number of injuries or fatalities per crash.

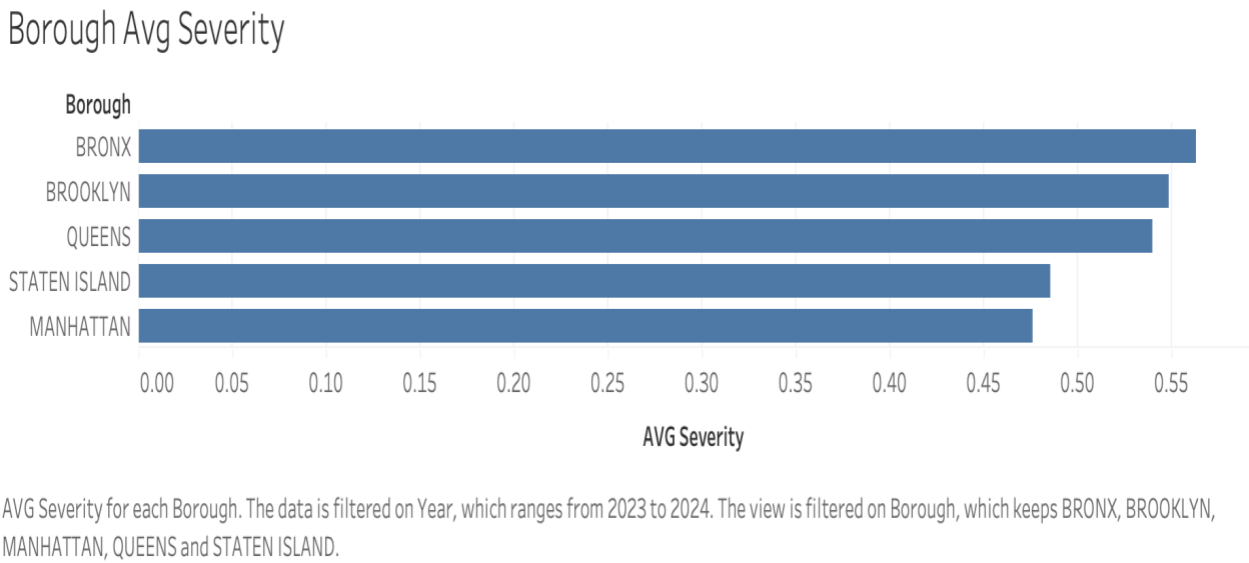
Geographic Crash Patterns

Figure 5 – Map of Crash Locations and Fatalities



The map visualization shows the geographic distribution of crashes across New York City. Circle size and color represent the number of fatalities associated with each crash. Areas with higher concentrations of crashes appear clustered around major roadways and densely populated boroughs.

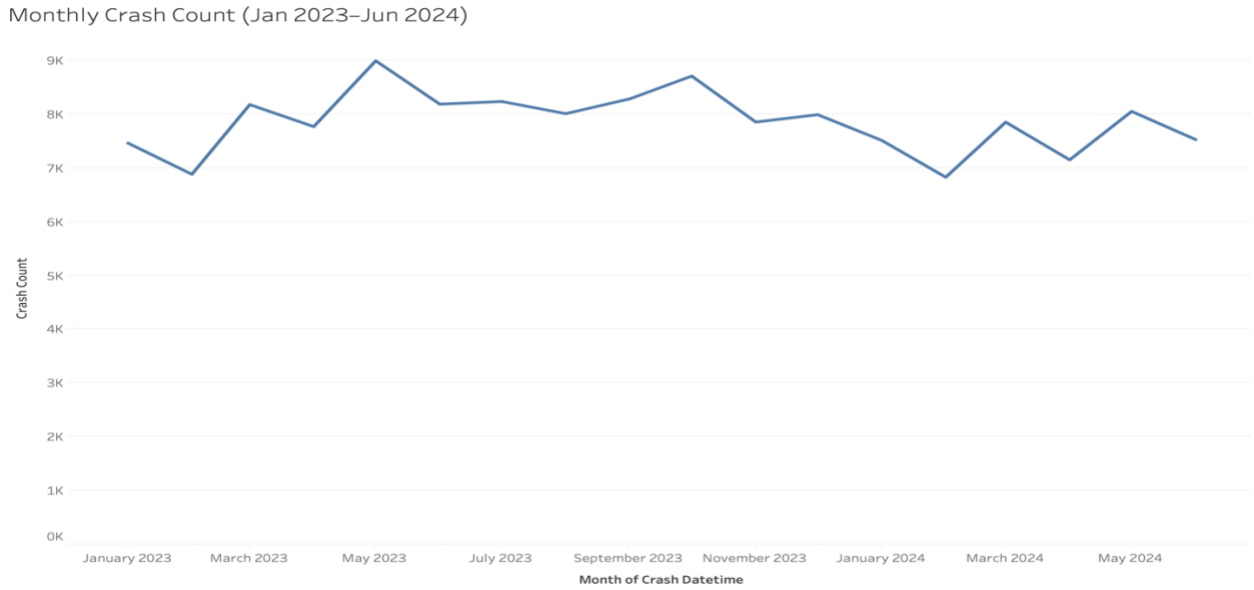
Figure 6 – Borough Average Crash Severity



This chart compares average crash severity across the five boroughs. The results show that the Bronx has the highest severity levels, followed by Brooklyn and Queens, while Manhattan has the lowest severity levels.

Temporal Crash Patterns

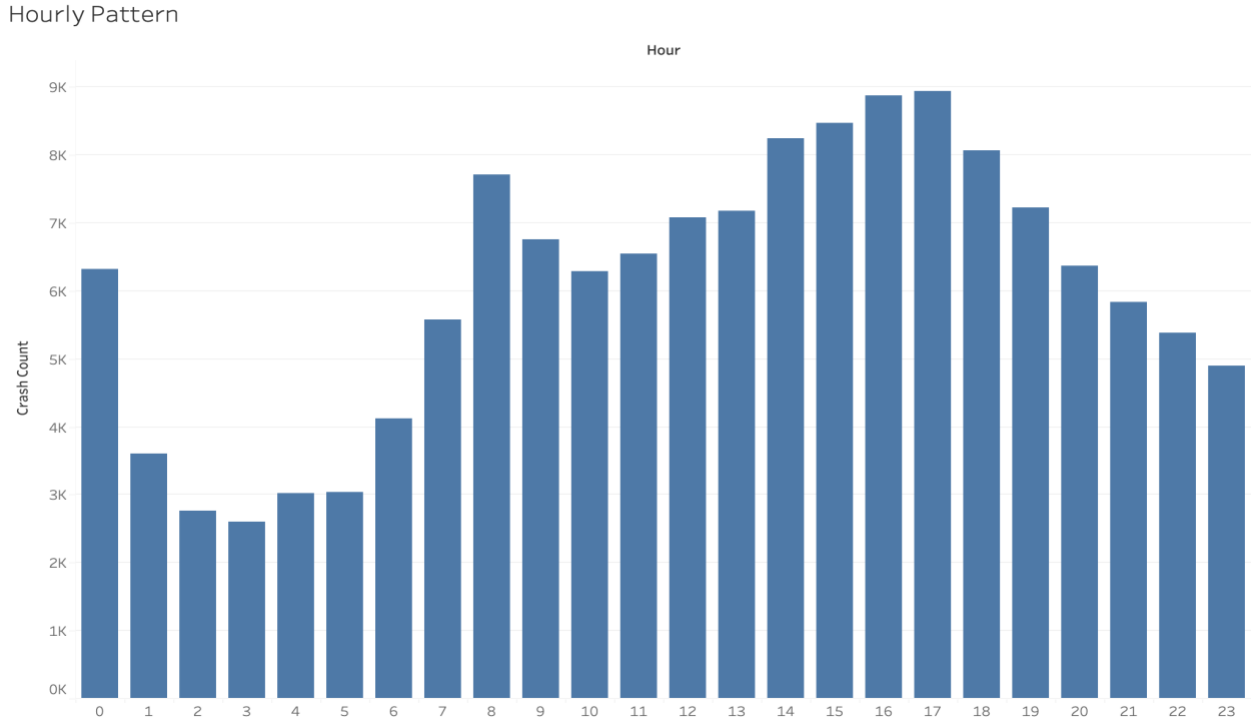
Figure 7 – Monthly Crash Count (Jan 2023–Jun 2024)



The trend of Crash Count for Crash Datetime Month. The data is filtered on Crash Datetime, which ranges from 1/1/2023 12:00:00 AM to 6/30/2024 12:30:00 PM.

This line chart illustrates the monthly trend of crash frequency. Crash counts remain relatively stable across the study period, generally ranging between 7,000 and 9,000 incidents per month.

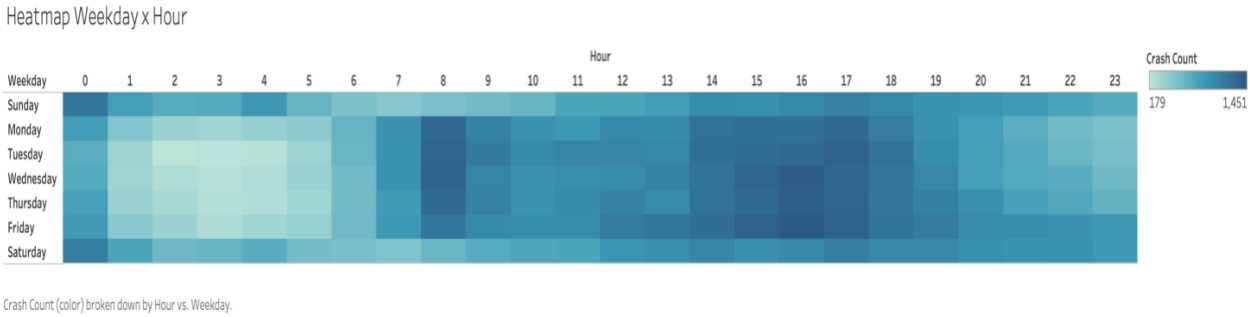
Figure 8 – Hourly Crash Pattern



Crash Count for each Hour.

The hourly crash pattern chart shows that crash frequency increases during daytime hours and peaks during late afternoon commuting periods.

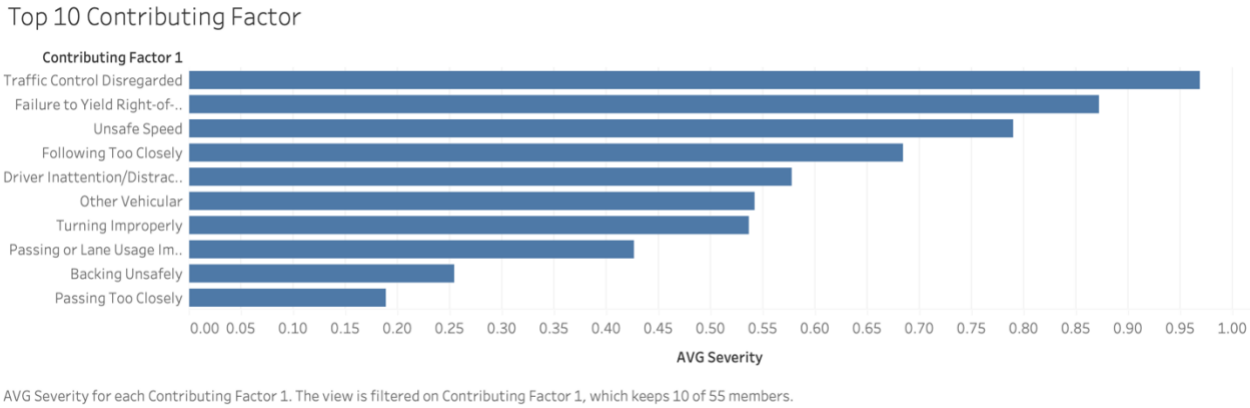
Figure 9 – Weekday by Hour Heatmap



This heatmap reveals that crash density is highest during weekday commuting hours, particularly in the morning and late afternoon.

Contributing Factors

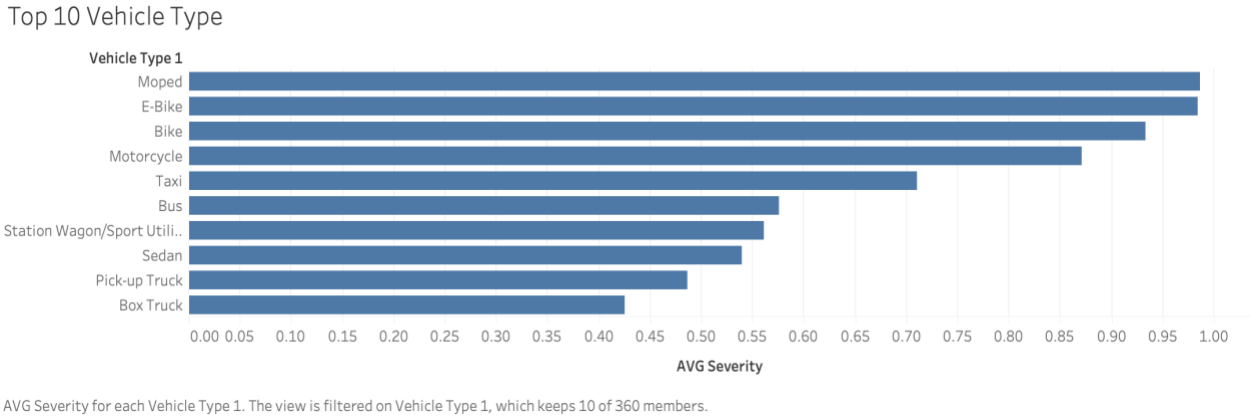
Figure 10 – Top Contributing Factors



Driver behavior plays a major role in crash severity. The most significant contributing factors include failure to yield right-of-way, unsafe speed, driver inattention, and following too closely.

Vehicle Types

Figure 11 – Top Vehicle Types in Severe Crashes



Vehicle types such as bicycles, mopeds, motorcycles, and e-bikes are associated with higher crash severity because these vehicles offer less protection compared to passenger cars.

Results Discussion

The visualizations generated in Tableau reveal several important patterns regarding crash severity in New York City.

First, the geographic distribution of crashes highlights the concentration of traffic incidents in densely populated urban areas. The map visualization indicates that crashes are heavily clustered around major roadways and intersections within boroughs such as the Bronx and Brooklyn. These areas often experience high traffic density and complex road networks, which increase the likelihood of collisions.

Additionally, borough-level analysis shows notable differences in crash severity. The Bronx exhibits the highest average crash severity among the five boroughs. This may be explained by several factors including roadway design, traffic congestion, and differences in transportation infrastructure. Brooklyn and Queens also display relatively high severity levels, while Manhattan shows lower severity levels despite its high population density. One possible explanation is that Manhattan's lower speed limits and extensive pedestrian infrastructure may reduce crash severity.

Temporal analysis reveals strong patterns associated with commuting behavior. Crash frequency increases during morning and evening commute periods, particularly between 3 PM and 6 PM. During these periods, traffic volume increases significantly as commuters travel between residential and commercial areas. Increased congestion, frequent lane changes, and driver impatience during rush hour contribute to higher collision risk.

The weekday-by-hour heatmap further illustrates the relationship between time and crash occurrence. Weekday mornings and afternoons show higher crash density compared to weekends. This pattern reflects work commuting patterns, as weekday traffic volume is significantly higher than weekend traffic.

Another important finding involves the role of driver behavior. The analysis of contributing factors demonstrates that driver-related violations account for many severe crashes. The most common factors include failure to yield right-of-way, unsafe speed, driver inattention, and disregarding traffic signals. These behaviors often occur when drivers fail to properly assess road conditions or engage in risky driving practices.

Vehicle type analysis reveals additional insights into crash severity. Vulnerable vehicle types such as bicycles, mopeds, and motorcycles are associated with higher severity outcomes. Unlike passenger vehicles, these transportation modes provide little physical protection during collisions, increasing the likelihood of injury.

These findings highlight the importance of both driver behavior and infrastructure design in traffic safety. Policies aimed at reducing distracted driving, enforcing speed limits, and improving bicycle infrastructure could significantly reduce crash severity in New York City.

Key Analytical Insights

Several important insights emerge from the data visualization analysis.

First, crash severity appears to be strongly influenced by traffic density and urban roadway complexity**.** Boroughs such as the Bronx and Brooklyn exhibit higher severity levels compared

to Manhattan. This pattern may be explained by roadway characteristics such as wider streets, higher speed limits, and greater interaction between local and through traffic.
Second, the temporal patterns suggest that commuting behavior significantly contributes to crash frequency. The concentration of crashes during afternoon rush hours indicates that increased traffic volume leads to more frequent vehicle interactions. Congestion during these periods may increase driver frustration and reduce reaction time, contributing to higher crash risk.
Third, driver behavior remains a dominant factor in crash severity. Contributing factors such as unsafe speed, failure to yield right-of-way, and driver distraction account for a large proportion of severe crashes. These findings are consistent with national traffic safety research indicating that human behavior plays a critical role in roadway safety.
Finally, the analysis highlights the vulnerability of non-passenger vehicle types. Motorcycles, bicycles, mopeds, and e-bikes are associated with higher injury severity due to their lack of protective structure. This finding emphasizes the importance of designing transportation systems that protect vulnerable road users.

Policy and Practical Implications

The findings from this study have several implications for transportation planning and public safety.
First, targeted enforcement of traffic laws could help reduce severe crashes. Law enforcement agencies could focus on high-risk behaviors such as speeding, distracted driving, and failure to yield. Increased enforcement during peak commuting hours may also help reduce collision rates.
Second, infrastructure improvements could significantly enhance road safety. For example, installing protected bike lanes can reduce interactions between cyclists and motor vehicles. Improved traffic signal timing and better intersection design may also help reduce collisions in high-risk areas.
Third, public awareness campaigns could educate drivers about the dangers of distracted driving and unsafe speed. Educational programs targeting high-risk driving behaviors may help promote safer driving practices.
Finally, urban planners could use crash data to identify high-risk locations and prioritize safety improvements. Data-driven planning allows cities to allocate resources more effectively and focus interventions where they are most needed.

Limitations

Although the analysis provides valuable insights, several limitations should be acknowledged.
First, the dataset does not include environmental variables such as weather conditions, which can significantly affect crash frequency. Rain, snow, and fog can reduce visibility and increase stopping distances, potentially contributing to crashes.
Second, the dataset does not contain detailed driver demographic information such as age, driving experience, or alcohol involvement. These factors are often important predictors of crash severity.
Third, the dataset relies on reported crash incidents. Minor collisions that are not reported to authorities may be missing from the data, which could introduce reporting bias.

Fourth, the analysis focuses primarily on descriptive statistics and visual patterns. Although these methods are effective for identifying trends, future research could incorporate predictive modeling techniques such as regression analysis or machine learning to better understand the relationships between variables.

Conclusion

This study analyzed motor vehicle crash severity patterns in New York City using data visualization techniques and Tableau dashboards. The analysis revealed several important patterns related to crash severity, including geographic differences across boroughs, temporal patterns related to commuting periods, and behavioral factors associated with severe crashes. The results highlight the importance of targeted traffic safety interventions, including improved enforcement of traffic laws, infrastructure improvements, and enhanced protections for vulnerable road users such as cyclists. By combining data analytics with visualization tools, this project demonstrates how large datasets can be transformed into meaningful insights that support better transportation policy decisions.

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